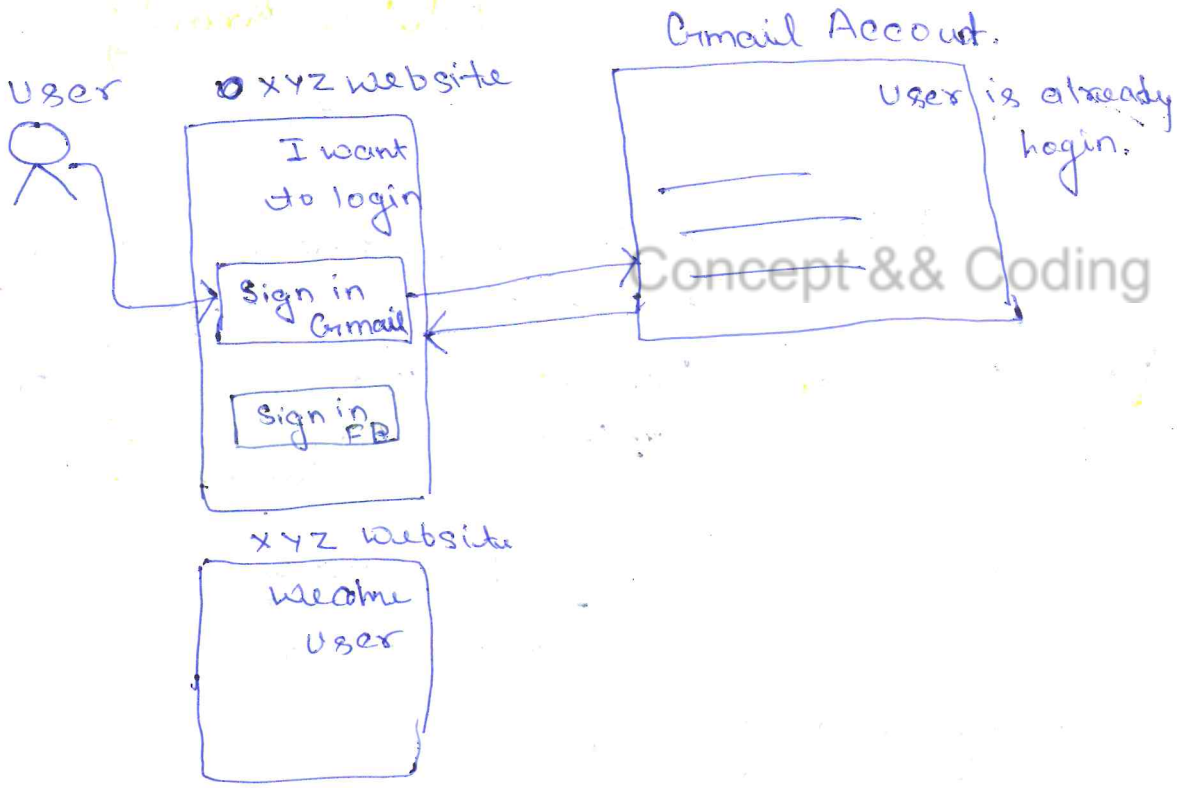
OAuth 2.0 (HLD)

OAuth stands for OPEN AUTHORIZATION

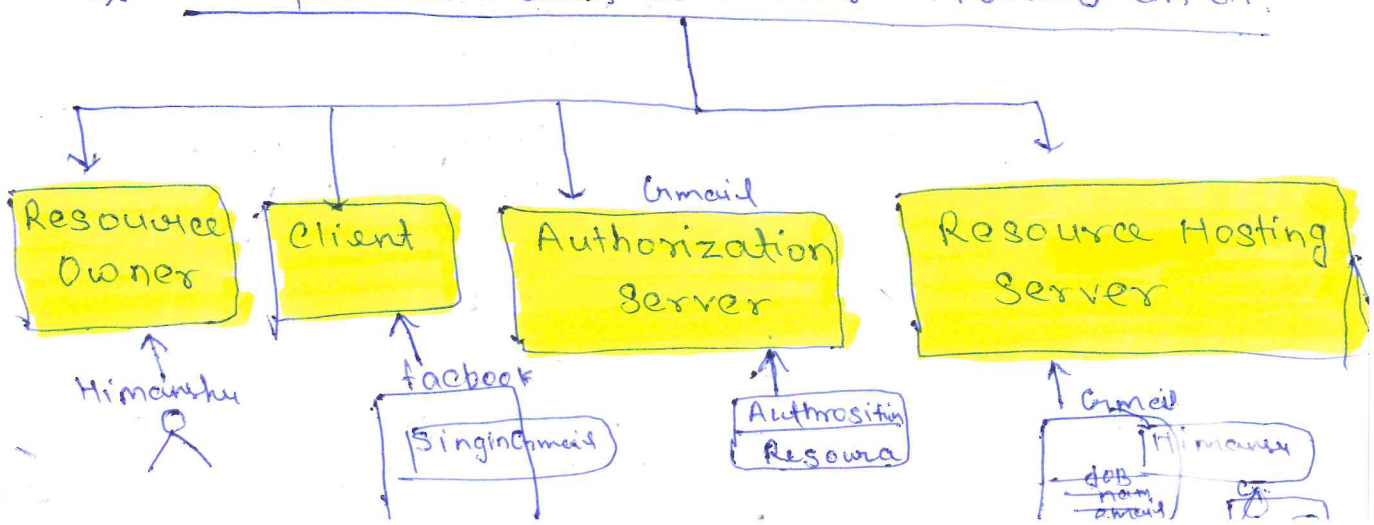
Why its used ?

- It's a Authorization framework
- Enables secure ~~thirst~~ third-party access to user protected data.



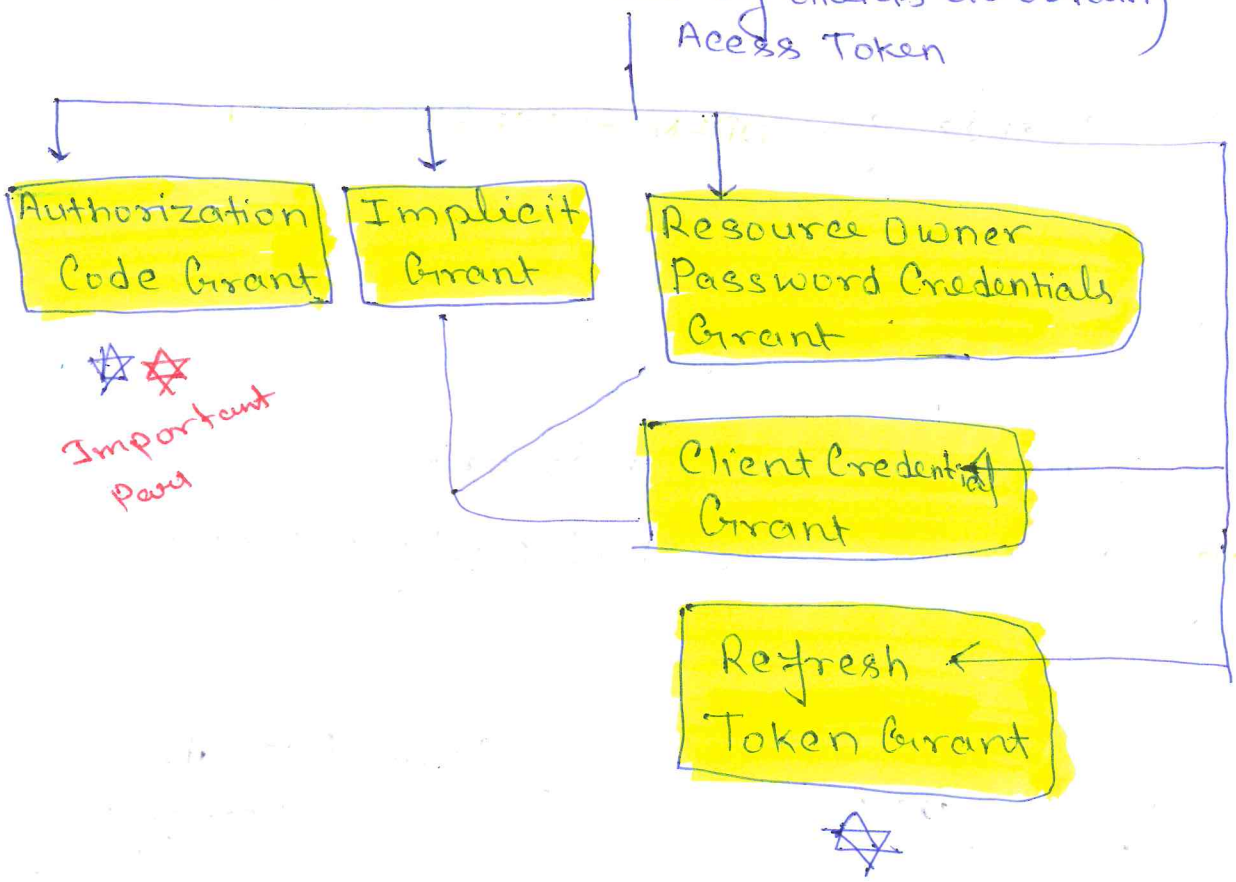
Concept && Coding

* 4 Important Roles or Actors involved in it:



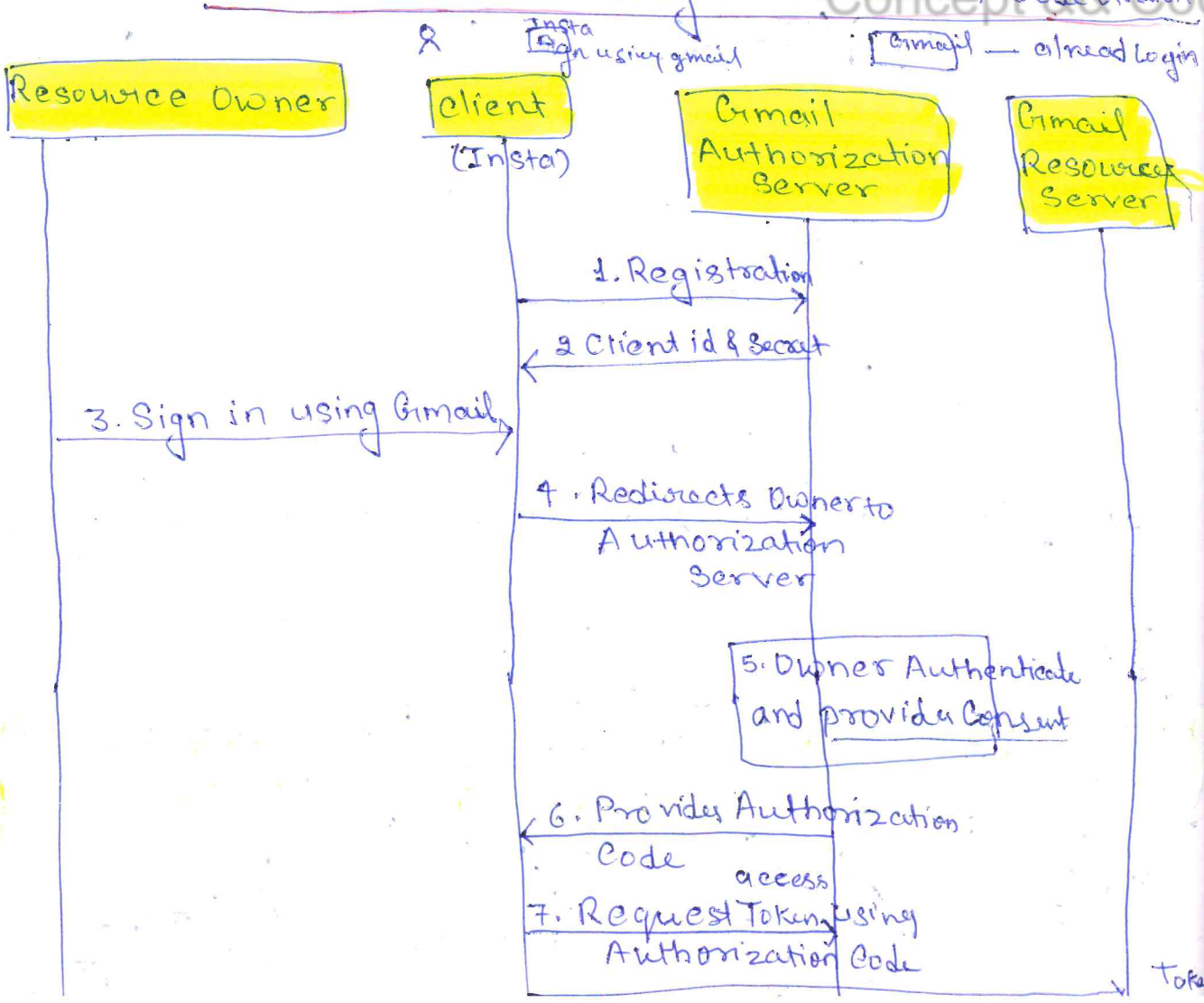
Authorization Grants types:

(Mechanism used by clients to obtain Acess Token)



End to End flow using Authorization Code Grant

Concept & Coding



10. Validate Token ←

11 Valid or Invalid →

12 ← 12. If valid Token, Provide the details else 400 error thrown

13 ← 13. Sign in successful or Failure.

API request and response:

1:- Registration Process:

Request:

POST/register

body:

```
{
  "client_name" : {my app name},
  "redirect_uris" ["https://mysample.com/",
                   "https://mywebsite.com/1"]
}
```

Instagram

3 URI

Response:

```
{
  "client_name" : {my app name},
  "client_id" : {unique id assigned to client}
  "client_secret: {confidential info, know only
                   to client and authorization
                   server, use for client authenti
```

Concept && Coding

Fetch Authorization Code:-

Request:

GET/authorize?
response_type = code
& client_id = {id which client gets during registration}
& redirect_uri = {callback URI like https:}
& scope = scope1 scope2 scope3
& state = {some random value like sj11}

Bearer:- is a security mechanism, and it means that client should add the TOKEN in the authorization header, whenever it want to access the protected resources.

Token value: It can either be JWT token or plain String.

Expiry Time: generally its in seconds. After expiry Refresh token should be used to get new Tokens.

Concept && Coding